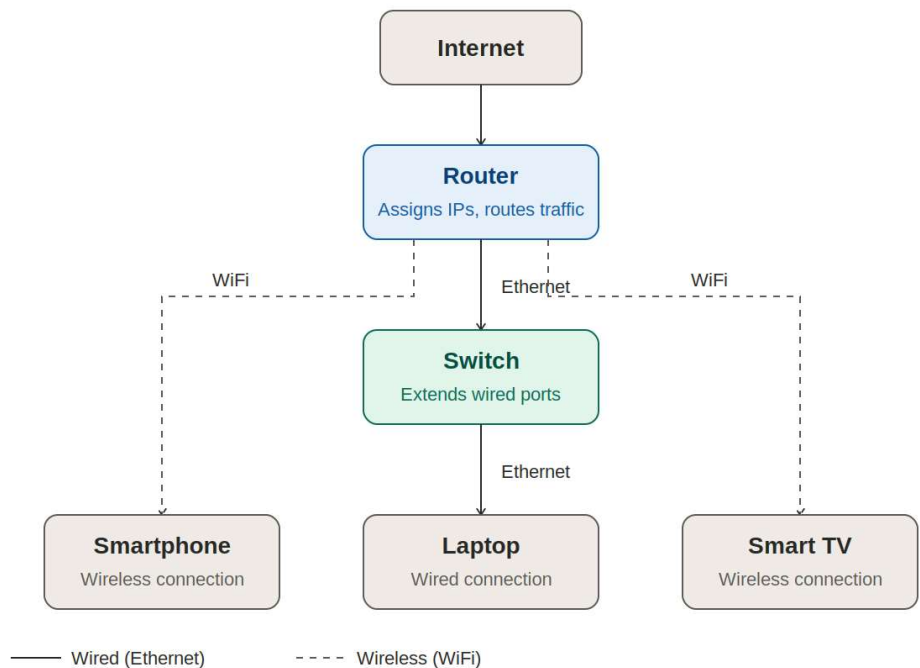


Task 1: Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.



The internet connects to the router, which links wirelessly (WiFi) to the smartphone and smart TV. The router also connects by Ethernet to a switch, which in turn provides a wired Ethernet connection to the laptop.

Task 2: Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily.

Device	Function	Real-world use case
Router	Connects your home network to the internet and assigns IP addresses to devices	Eg: Jio Fiber/Airtel router lets every device in the house - phone, laptop, TV - get online at once, so you can stream YouTube on the TV while browsing on your phone
Switch	Connects multiple wired devices within the same local network, expanding available Ethernet ports	Useful for LAN gaming - connecting a desktop PC and gaming console via Ethernet through a switch gives lower latency than WiFi
WiFi range extender / repeater	Picks up an existing WiFi signal and rebroadcasts it to cover a dead zone.	It is used to cover a dead zone, without needing a wired connection back to the router (unlike an access point, which usually does need a wired uplink). Handy if any room has a weak WIFI signal.

Access point	Extends WiFi coverage by broadcasting a wireless signal, often used when the router's signal doesn't reach far enough	Connects to the router via a wired Ethernet cable (usually run through the wall or along the ceiling). Placing a WiFi access point on a different floor so video calls on Zoom don't drop when you move away from the main router
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Task 3: Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved.

When you open Zomato and tap "Place order." The phone's WiFi radio sends that request directly to the router, which picks up the wireless signal and converts it into data the wired network understands.

The router checks the destination address, replaces your phone's private IP with your home's public IP (a process called NAT), and sends the request out through your ISP to Zomato's servers.

The response — order confirmation, restaurant details — travels back along the exact same path in reverse: internet → router → your phone, arriving as the "Order placed" screen.

Task 4 : Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.

- Ethernet cable: A single Ethernet cable (Cat5e or Cat6) from a port on the router all the way to the gaming PC, plugging directly into the PC's Ethernet port at the other end. No adapters, no signal conversion, It is Fastest and most stable and no signal interference.
- Powerline adapter: One unit plugs into a wall socket near the router and connects to it via Ethernet cable. A second unit plugs into a wall socket near the gaming PC and connects to it via Ethernet. The pair uses the home's existing electrical wiring to carry the network signal between rooms, so no new cable needs to be run.
- WiFi extender with an Ethernet port (wireless bridge): The extender is placed roughly halfway between the router and the PC, where it still gets a decent signal. It picks up the router's WiFi wirelessly, then provides a wired Ethernet port that plugs straight into the PC's Ethernet port, effectively translating WiFi back into a wired connection.